Compulsory Assignment 1

Predictive Analytics (KAN-CDSCO1005U)

Alejandro Gato (161989), Alexander Myrup (160363), Pontus Neumann (185912), Linus Yu (160714)

2026-03-01

Question 1: GDP Level – Plot and ACF

```r
#install.packages("AER")
#install.packages("fpp3")
#install.packages("MetBrewer")

library(AER)
library(fpp3)
library(ggplot2)
library(MetBrewer)
```

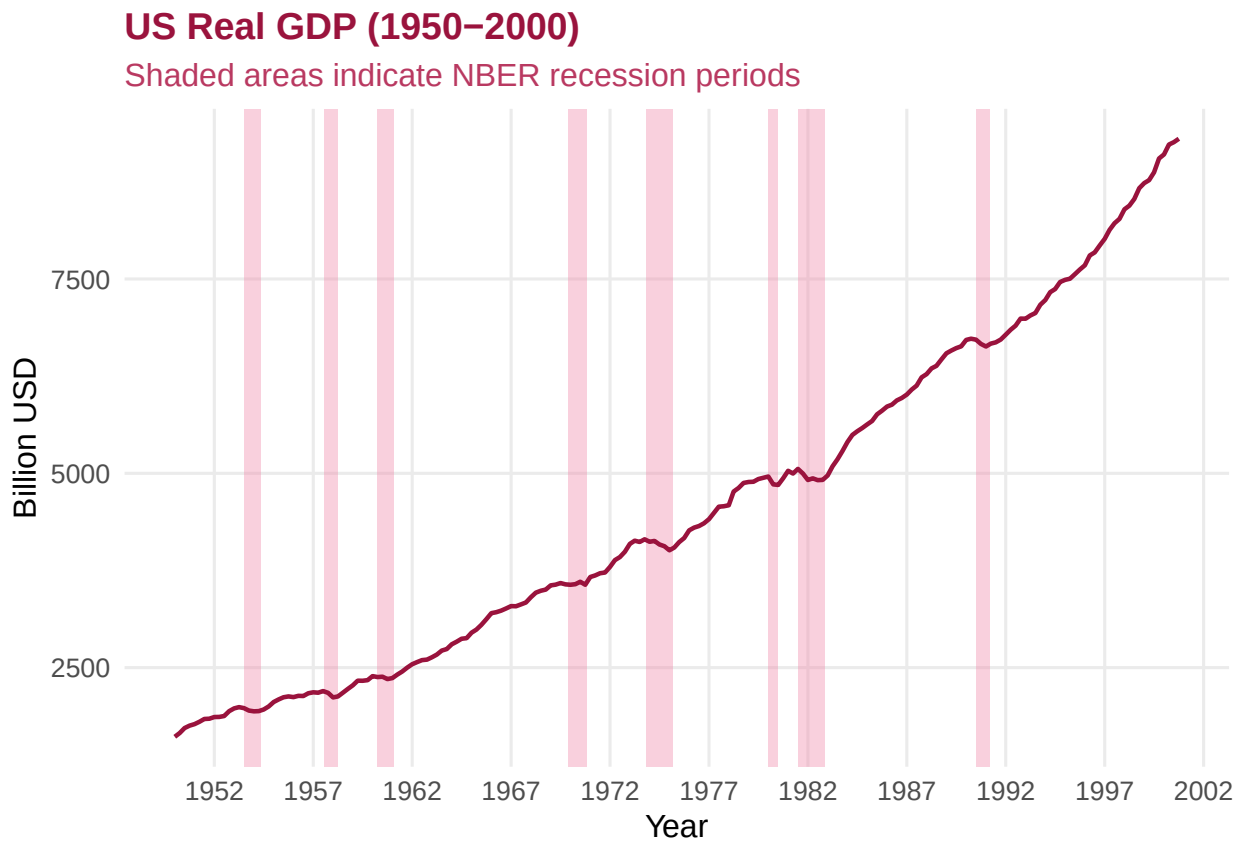
```r
data("USMacroG", package = "AER")
gdp <- USMacroG[, "gdp"]

# Load full Benedictus palette upfront for use throughout the assignment
bene <- met.brewer("Benedictus")

# Convert GDP to data frame for ggplot
gdp_df <- data.frame(
  date = as.Date(time(gdp)),
  gdp  = as.numeric(gdp)
)

# NBER recession periods within sample (1950-2000)
recessions <- data.frame(
  start = as.Date(c("1953-07-01", "1957-08-01", "1960-04-01", "1969-12-01",
                    "1973-11-01", "1980-01-01", "1981-07-01", "1990-07-01")),
  end   = as.Date(c("1954-05-01", "1958-04-01", "1961-02-01", "1970-11-01",
                    "1975-03-01", "1980-07-01", "1982-11-01", "1991-03-01"))
)
```

```r
ggplot(gdp_df, aes(x = date, y = gdp)) +
  geom_rect(data = recessions,
            aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf),
            inherit.aes = FALSE,
            fill = bene[4], alpha = 0.4) +
  geom_line(color = bene[1], linewidth = 0.8) +
  scale_x_date(date_breaks = "5 years", date_labels = "%Y") +
  labs(
    title    = "US Real GDP (1950-2000)",
    subtitle = "Shaded areas indicate NBER recession periods",
    x        = "Year",
    y        = "Billion USD"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title       = element_text(color = bene[1], face = "bold"),
    plot.subtitle    = element_text(color = bene[2]),
    panel.grid.minor = element_blank()
  )
```

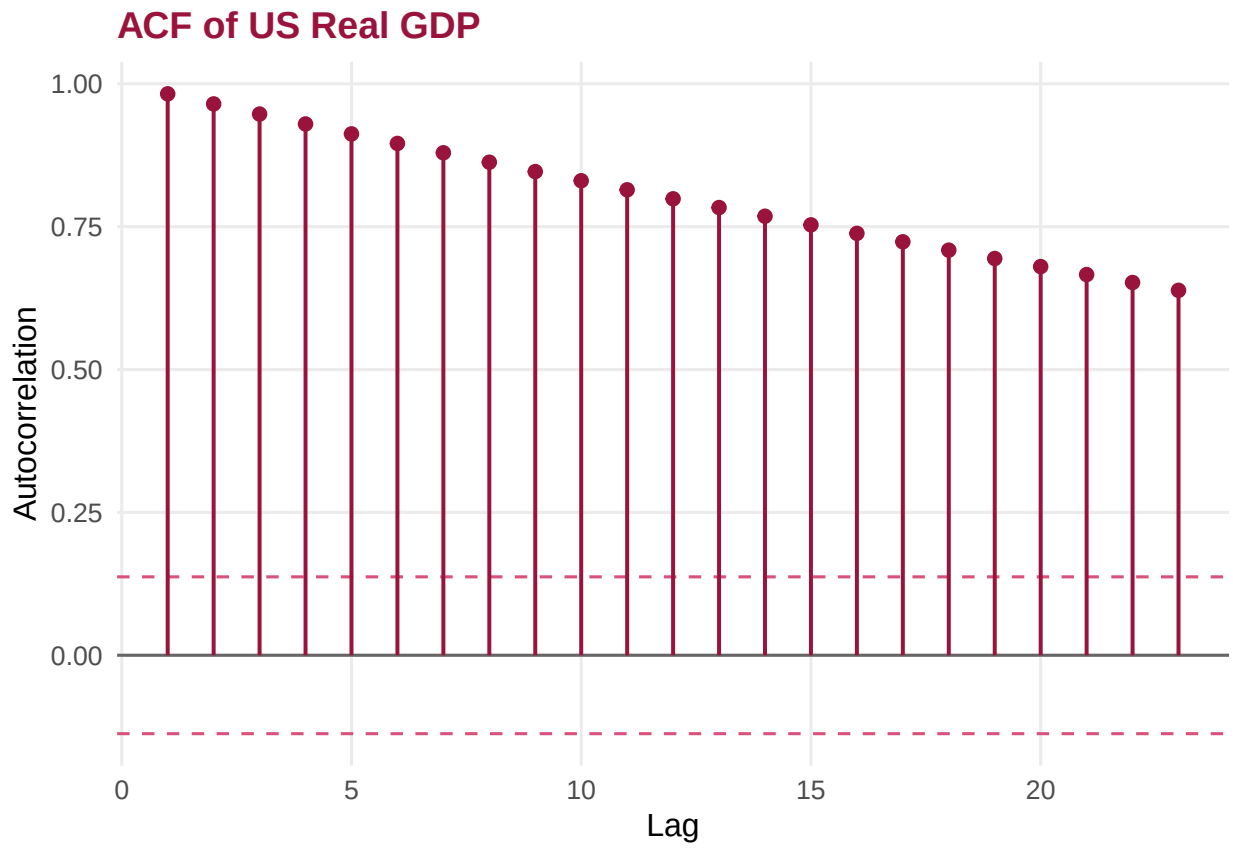


The GDP series displays a clear upward trend throughout the entire sample period from 1950 to 2000, indicating that the series is non-stationary in its level. There is no visible sign of seasonality, though brief slowdowns in growth are visible around the shaded recession periods, most notably in the early 1970s and early 1980s. The variance also appears to increase over time alongside the level of the series, which suggests that a log transformation may be appropriate before modelling.

```
acf_data <- acf(as.numeric(gdp), plot = FALSE)
acf_df <- data.frame(
  lag = as.numeric(acf_data$lag[-1]),
  acf = as.numeric(acf_data$acf[-1])
)
ci <- qnorm(0.975) / sqrt(length(gdp))

ggplot(acf_df, aes(x = lag, y = acf)) +
  geom_hline(yintercept = 0, color = "grey40") +
  geom_hline(yintercept = c(-ci, ci), linetype = "dashed", color = bene[3]) +
  geom_segment(aes(xend = lag, yend = 0), color = bene[1], linewidth = 0.7) +
  geom_point(color = bene[1], size = 2) +
  labs(
    title = "ACF of US Real GDP",
    x     = "Lag",
    y     = "Autocorrelation"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title       = element_text(color = bene[1], face = "bold"),
```

```
    panel.grid.minor = element_blank()
  )
```



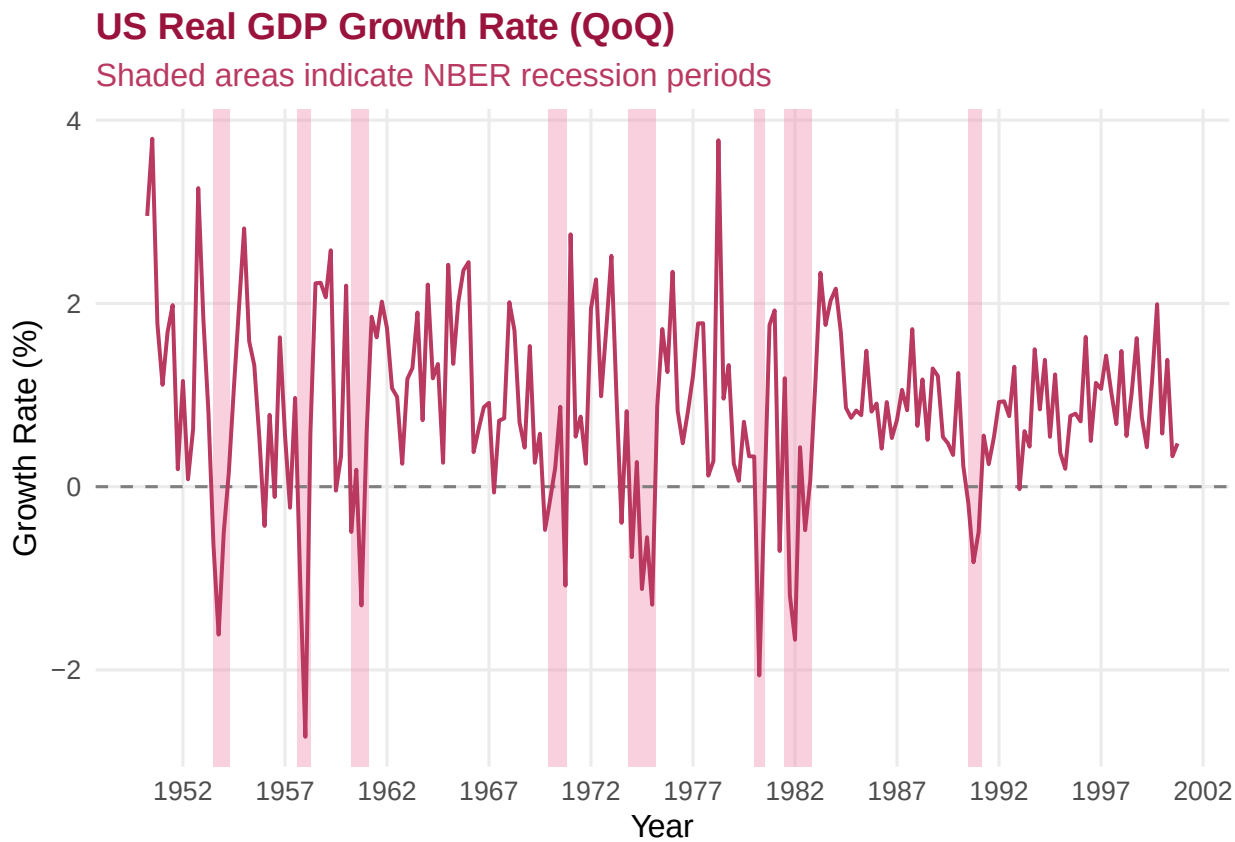
The ACF shows that GDP values are very closely related to their recent past. The bars start near 1.0 and only decrease gradually. Even at a lag of 20 quarters (5 years), the correlation is still well above the dashed significance lines. This slow, steady decline is exactly what we'd expect from a series with a strong upward trend: because GDP keeps rising, any given quarter naturally resembles the quarters before it. This confirms that the raw GDP series is non-stationary, highly persistent and needs to be transformed before we can model it meaningfully.

Question 2: GDP Growth Rate – Transformation, Plot, ACF, and Ljung-Box Test

```r
gdp_growth <- diff(log(gdp)) * 100  # Quarter-on-quarter log growth rate in %

gdp_growth_df <- data.frame(
  date   = as.Date(time(gdp_growth)),
  growth = as.numeric(gdp_growth)
)
```

```r
ggplot(gdp_growth_df, aes(x = date, y = growth)) +
  geom_rect(data = recessions,
            aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf),
            inherit.aes = FALSE,
            fill = bene[4], alpha = 0.4) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey50") +
  geom_line(color = bene[2], linewidth = 0.7) +
  scale_x_date(date_breaks = "5 years", date_labels = "%Y") +
  labs(
    title    = "US Real GDP Growth Rate (QoQ)",
    subtitle = "Shaded areas indicate NBER recession periods",
    x        = "Year",
    y        = "Growth Rate (%)"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title       = element_text(color = bene[1], face = "bold"),
    plot.subtitle    = element_text(color = bene[2]),
    panel.grid.minor = element_blank()
  )
```

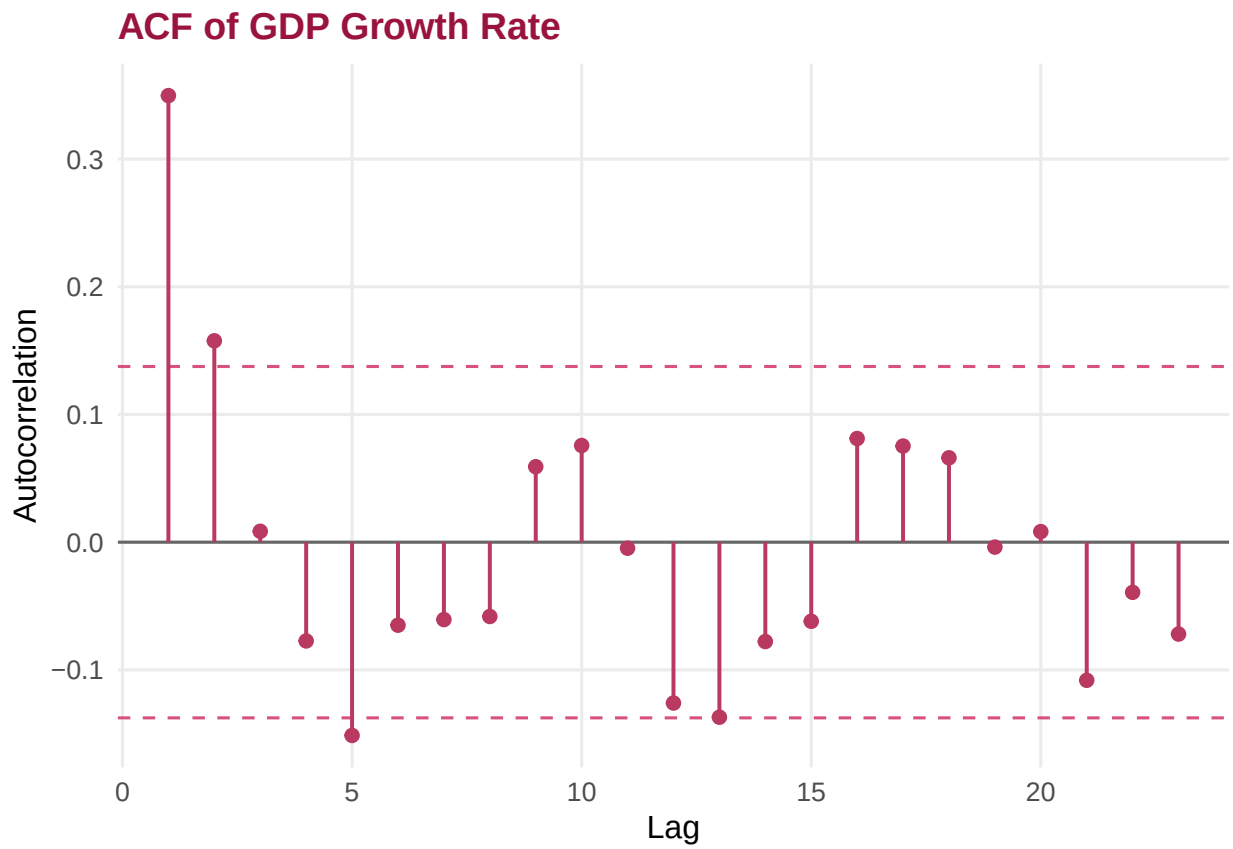


After taking the log-difference, the series no longer exhibits a trend – it fluctuates around a roughly constant mean close to zero, suggesting that differencing has successfully removed the non-stationarity present in the level series. There is no clear seasonal pattern, though some cyclical variation is visible, with more volatile and negative growth periods coinciding with the shaded recession episodes, most notably the sharp contractions of the early 1970s oil shock and the double-dip recession of 1980-1982. The variance appears broadly stable across the sample, supporting the appropriateness of the log transformation applied.

```
acf_growth <- acf(as.numeric(gdp_growth), plot = FALSE)
acf_growth_df <- data.frame(
  lag = as.numeric(acf_growth$lag[-1]),
  acf = as.numeric(acf_growth$acf[-1])
)
ci_growth <- qnorm(0.975) / sqrt(length(gdp_growth))

ggplot(acf_growth_df, aes(x = lag, y = acf)) +
  geom_hline(yintercept = 0, color = "grey40") +
  geom_hline(yintercept = c(-ci_growth, ci_growth),
             linetype = "dashed", color = bene[3]) +
  geom_segment(aes(xend = lag, yend = 0), color = bene[2], linewidth = 0.7) +
  geom_point(color = bene[2], size = 2) +
  labs(
    title = "ACF of GDP Growth Rate",
    x     = "Lag",
    y     = "Autocorrelation"
  ) +
  theme_minimal(base_size = 12) +
```

```r
  theme(
    plot.title       = element_text(color = bene[1], face = "bold"),
    panel.grid.minor = element_blank()
  )
```



The ACF of the growth rate decays much more rapidly than that of the level series, consistent with a stationary process. Lag 1 is clearly significant, lag 2 is borderline, and lag 5 is significant on the negative side. Overall, most lags fall within the confidence bands, indicating the growth rate behaves much closer to white noise than the original GDP series.

```r
lb_test <- Box.test(gdp_growth, lag = 12, type = "Ljung-Box")

lb_tbl <- data.frame(
  Test = "Ljung-Box",
  Lags = 12,
  Statistic = round(unname(lb_test$statistic), 3),
  df        = unname(lb_test$parameter),
  p.value   = format.pval(lb_test$p.value, digits = 3, eps = 1e-3),
  check.names = FALSE
)

knitr::kable(
  lb_tbl,
  col.names = c("Test", "Lags", "X-squared", "df", "p-value"),
  caption   = "Ljung-Box test for GDP growth rate (H0: no autocorrelation up to lag 12).",
```

```
  align     = "lcccc"
)
```

Table 1: Ljung-Box test for GDP growth rate (H0: no autocorrelation up to lag 12).

Test	Lags	X-squared	df	p-value
Ljung-Box	12	44.262	12	<0.001

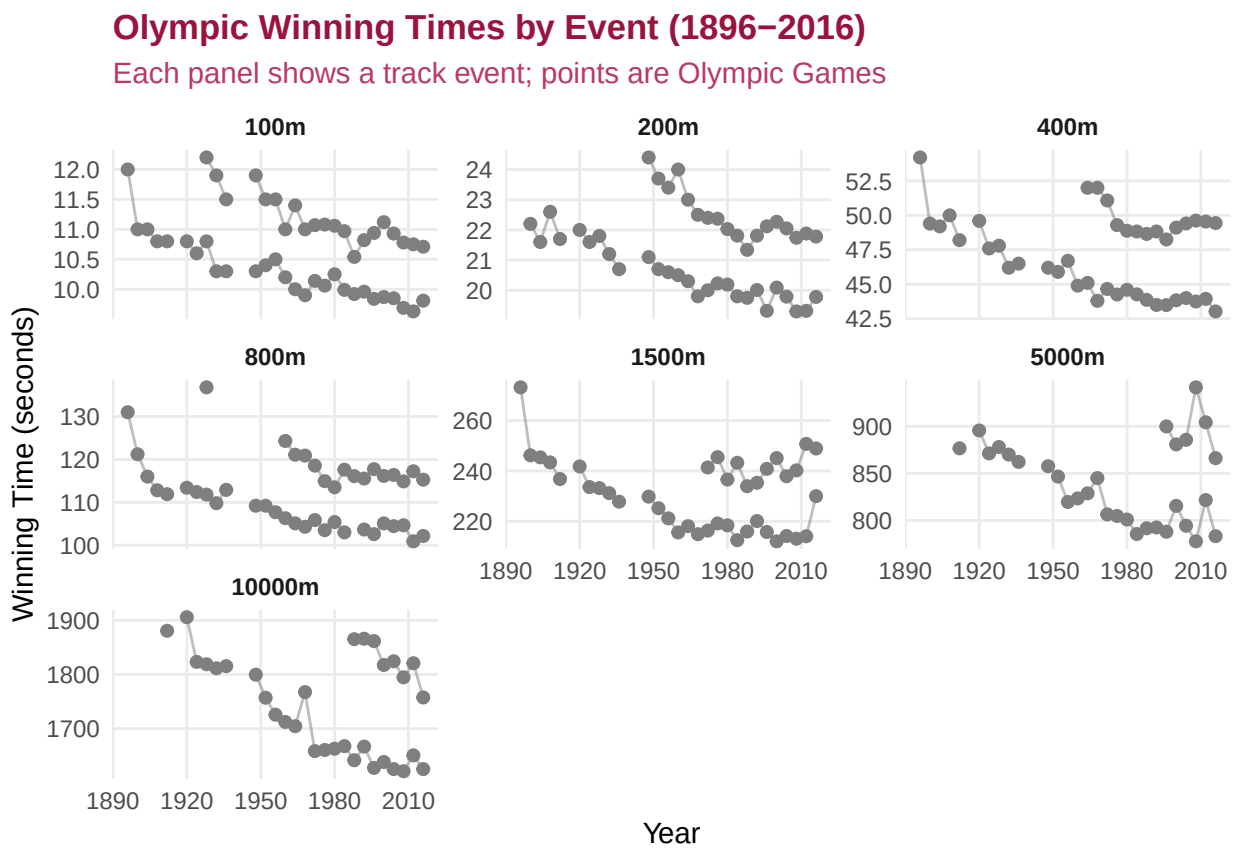
The Ljung-Box test evaluates the null hypothesis that the first 12 autocorrelations are jointly equal to zero, i.e. that the series is white noise. The test statistic is 44.262 with a p-value of <0.001. Since the p-value is below the conventional 5% significance level, we reject the null hypothesis and conclude that statistically significant autocorrelation is present in the GDP growth rate series. This finding is consistent with the ACF plot and implies that the growth rate is not pure white noise, which has important implications for forecasting – a model that exploits this autocorrelation structure should outperform a naive no-change forecast.

Question 3: Olympic Running

```r
data <- olympic_running
head(data)
```

```
## # A tsibble: 6 x 4 [4Y]
## # Key:       Length, Sex [1]
##    Year Length Sex    Time
##   <int>  <int> <chr> <dbl>
## 1  1896    100 men    12
## 2  1900    100 men    11
## 3  1904    100 men    11
## 4  1908    100 men    10.8
## 5  1912    100 men    10.8
## 6  1916    100 men    NA
```

```r
ggplot(data, aes(x = Year, y = Time, color = Sex)) +
  geom_point(size = 1.8) +
  geom_line(alpha = 0.5) +
  facet_wrap(~ Length, scales = "free_y", ncol = 3,
             labeller = labeller(Length = function(x) paste0(x, "m"))) +
  scale_color_manual(values = c("Men" = bene[1], "Women" = bene[5])) +
  labs(
    title    = "Olympic Winning Times by Event (1896-2016)",
    subtitle = "Each panel shows a track event; points are Olympic Games",
    x        = "Year",
    y        = "Winning Time (seconds)",
    color    = NULL
  ) +
  theme_minimal(base_size = 11) +
  theme(
    plot.title       = element_text(color = bene[1], face = "bold"),
    plot.subtitle    = element_text(color = bene[2]),
    panel.grid.minor = element_blank(),
    legend.position  = "bottom",
    strip.text       = element_text(face = "bold")
  )
```



The plots reveal several consistent features across all events. First, there is a clear downward trend in winning times over the course of the twentieth century, reflecting improvements in athletic training, nutrition, equipment, and race strategy. Second, men consistently outperform women across all distances, though women's events were introduced later and show steeper initial declines as the sport professionalised. Third, the rate of improvement appears to slow in more recent decades, suggesting diminishing returns as athletes approach physiological limits. Some events also show notable outliers, potentially reflecting unusual race conditions, doping disqualifications, or wars causing gaps in the data (1916, 1940, 1944).

```
data <- as.data.frame(olympic_running)

  fits <- data %>%
    group_by(Length, Sex) %>%
    filter(!is.na(Time)) %>%
    summarise(
      slope = coef(lm(Time ~ Year))[2],
      .groups = "drop"
    )

  fits_tbl <- fits %>%
    mutate(`Avg. change per year (sec)` = sprintf("%.4f", slope)) %>%
    select(Event = Length, Sex, `Avg. change per year (sec)`)

  knitr::kable(
    fits_tbl,
    caption = "Estimated average change in winning time per year for each Olympic event.",
    align = "llc"
```

```
  )
```

Table 2: Estimated average change in winning time per year for each Olympic event.

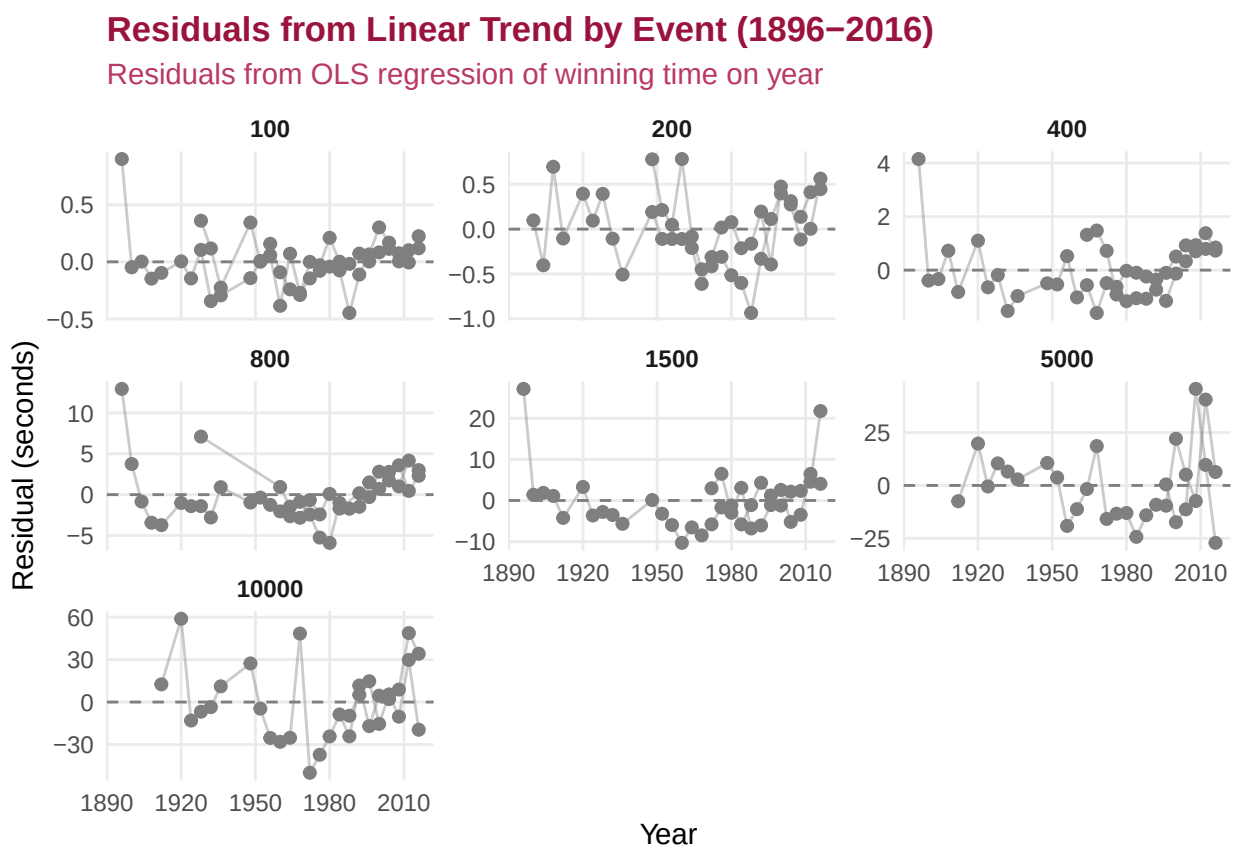
Event	Sex	Avg. change per year (sec)
100	men	-0.0126
100	women	-0.0142
200	men	-0.0249
200	women	-0.0336
400	men	-0.0646
400	women	-0.0401
800	men	-0.1518
800	women	-0.1980
1500	men	-0.3151
1500	women	0.1468
5000	men	-1.0299
5000	women	-0.3030
10000	men	-2.6659
10000	women	-3.4962

All slopes are negative except Women's 1500m, confirming that winning times have generally decreased over the period. Longer events show larger absolute decreases per year (e.g. Men's 10,000m at -2.67 sec/year vs Men's 100m at -0.01 sec/year), which is expected since longer races have more total time available for improvement. Women's events tend to show steeper declines than men's in the same event, reflecting the later professionalisation of women's athletics and thus larger room for early gains. The anomalous positive slope for Women's 1500m (+0.15 sec/year) is likely an artefact of the small sample and the event's late introduction to the Olympics, making the linear fit unreliable for that group.

```r
data <- data %>%
  group_by(Length, Sex) %>%
  filter(!is.na(Time)) %>%
  mutate(
    fitted   = lm(Time ~ Year)$fitted.values,
    residual = lm(Time ~ Year)$residuals
  ) %>%
  ungroup()

ggplot(data, aes(x = Year, y = residual, color = Sex)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey50") +
  geom_point(size = 1.8) +
  geom_line(alpha = 0.4) +
  facet_wrap(~ Length, scales = "free_y", ncol = 3) +
  scale_color_manual(values = c("Men" = bene[1], "Women" = bene[5])) +
  labs(
    title    = "Residuals from Linear Trend by Event (1896-2016)",
    subtitle = "Residuals from OLS regression of winning time on year",
    x        = "Year",
    y        = "Residual (seconds)",
    color    = NULL
  ) +
  theme_minimal(base_size = 11) +
```

```r
  theme(
    plot.title       = element_text(color = bene[1], face = "bold"),
    plot.subtitle    = element_text(color = bene[2]),
    panel.grid.minor = element_blank(),
    legend.position  = "bottom",
    strip.text       = element_text(face = "bold")
  )
```



The residual plots show a clear non-random pattern across most events. Residuals tend to be positive in the early years, negative in the middle of the sample, and positive again in recent decades. This U-shaped pattern indicates that winning times improved rapidly in the early-to-mid twentieth century and have since slowed, which a straight line cannot capture. This suggests a nonlinear trend would better describe the data. Some events also show clusters of large residuals around the World War years, reflecting gaps in the Olympic schedule.

```r
predictions <- data %>%
  filter(!is.na(Time)) %>%
  group_by(Length, Sex) %>%
  group_modify(~ {
    model <- lm(Time ~ Year, data = .x)
    pred  <- predict(model,
                     newdata  = data.frame(Year = 2020),
                     interval = "prediction",
                     level    = 0.95)
    data.frame(
      Predicted = round(pred[1, "fit"], 2),
```

```r
      Lower     = round(pred[1, "lwr"], 2),
      Upper     = round(pred[1, "upr"], 2)
    )
  }) %>%
  ungroup()

knitr::kable(
  predictions,
  col.names = c("Event", "Sex", "Predicted (sec)", "Lower 95% PI", "Upper 95% PI"),
  caption   = "Predicted winning times for the 2020 Olympics with 95% prediction intervals.",
  align     = "llccc"
)
```

Table 3: Predicted winning times for the 2020 Olympics with 95% prediction intervals.

Event	Sex	Predicted (sec)	Lower 95% PI	Upper 95% PI
100	men	9.53	9.03	10.04
100	women	10.53	10.02	11.04
200	men	19.12	18.39	19.85
200	women	21.20	20.02	22.38
400	men	42.04	39.55	44.53
400	women	48.44	45.87	51.01
800	men	99.24	91.78	106.70
800	women	111.48	103.41	119.56
1500	men	207.00	189.27	224.73
1500	women	245.46	232.19	258.72
5000	men	772.82	737.81	807.84
5000	women	892.11	782.40	1001.83
10000	men	1580.35	1515.75	1644.95
10000	women	1763.02	1706.69	1819.35

The predictions provide point estimates and 95% prediction intervals for the winning time in each event at the 2020 Tokyo Olympics. These forecasts rest on several important assumptions. First, the linear trend is assumed to continue into the future at the same constant rate observed historically – an assumption that is likely violated, as the residual analysis above suggests the rate of improvement has been slowing. This means the model will tend to produce optimistic (too fast) predictions, particularly for events where the curvature in residuals is most pronounced. Second, the model assumes that no structural breaks occur – such as new technological innovations, rule changes, or doping scandals – that would shift the trend. Third, the prediction intervals are derived under the assumption of normally distributed, homoskedastic residuals; if this does not hold, the intervals may be too narrow or too wide. In practice, the actual 2020 results can be compared against these predictions to evaluate the model's out-of-sample performance.